Institute/ School Name	School of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Program Name	Bachelor of Engineering (Computer Science & Engineering): B.E (CSE)		
Course Code	25CS029	Course Name	Advanced Algorithm Design & Implementation
L-T-P (Per Week)	3-0-4	Course Credits	05
Academic Year	2025-26	Semester/Batch	6th /2023-2027
Course Coordinator	Dr. Osho Sharma		

1. Course Outline:

Memory models, DMA, recursion, linked lists, stacks, priority queues, hashing, trees, string operations, sliding windows, bit manipulation, OOP concepts and implementation, divide-and-conquer algorithms, greedy algorithms, advanced problems on arrays, dynamic programming, and graph algorithms, all with considerations for professional ethics and sustainability.

2. Programme Outcomes (POs):

At the end of the programme, students will be able to achieve knowledge about the following:	
PO 1	**Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning**: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3. Course Learning Outcomes (CLO):

After completing the course, the students will be able to:

CLO1: Develop strong object-oriented programming skills to understand OOP concepts and create class hierarchies using an object-oriented design process.
CLO2: Apply algorithmic skills to implement and analyze trees and graph traversal techniques.
CLO3: Demonstrate problem-solving skills by implementing efficient searching and sorting techniques.
CLO4: Acquire practical programming skills to explore and implement stack, queue, and linked list operations.
CLO5: Foster innovation and employability skills by solving real-world problems using advanced algorithms such as Greedy, Backtracking, and Dynamic Programming.

4. CLO-PO Mapping Matrix:

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1	M			L								M
CLO2		L	H		M							
CLO3		L			M		L					M
CLO4	M			L					M			
CLO5			L		M							M

5. ERISE Grid Mapping:

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	1
Research/Innovation	3
Skills	5
Employability	2

6. Recommended Books (Reference Books/Text Books):

B01: Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. Introduction to algorithms (4th Edition). MIT press.
B02: McLaughlin, B., Pollice, G., & West, D. Head First Object-Oriented Analysis and Design: A Brain Friendly Guide to OOA&D (1st Edition). O'Reilly Media, Inc.
B03: Jagtap, A. M., & Mali, A. S. Data Structures Using C: A Practical Approach for Beginners (1st Edition). Chapman and Hall/CRC.
B04: West, D. B. Introduction to graph theory (1st Edition). Upper Saddle River: Prentice hall.
B05: Laaksonen, A. Guide to competitive programming (2nd Edition). Springer.

7. Other readings and relevant websites:

Resources	Link of Journals, Magazines, Websites and Research Papers
R1	https://jeffe.cs.illinois.edu/teaching/algorithms/
R2	https://stmjournals.in/international-journal-of-data-structures/
R3	https://www.codemag.com/Article/2111021/Data-Structures-and-Algorithms
R4	https://www.researchgate.net/publication/329390711_Dynamic_programming
Resources	**Link of Audio-Video resources**
V1	https://www.youtube.com/watch?v=-QcPo_DWJk4
V2	https://www.youtube.com/watch?v=iTMn0Kt18tg
V3	https://www.youtube.com/watch?v=r4-cftqTcdI
V4	https://www.youtube.com/watch?v=RwBCrVB98s8
V5	https://www.youtube.com/watch?v=PVrbRrI0jG4

* Resources uploaded on ERP system is accessible to all the students registered for the course.

8. Recommended Tools and Platforms:

- Visual Studio Code
- Code Quotient (online platform-https://codequotient.com)

9. Course Plan:

Lecture Number	Topics	Weightage in ETE (%)	Instructional Resources
1-4	Memory Model, Dynamic Memory Allocation	30	B01, B05, R2, R3
5-9	Design and Analysis of Recursive and Backtracking Algorithms		
9-14	Advance problems based on Linked List		
14-18	Advance problems based on Stack		
18-23	Sliding Window Technique		
23-28	String Operations, Pattern Matching: KMP Algorithm, Z-Value		
29-33	Rabin Karp Algorithm		
34-37	Operations on Binary Tree, Binary Tree Traversal Algorithms		
38-42	Binary Search Tree Operation and Traversals, Problems based on BST		
43-46	Introduction to Hashing Technique, Problems based on Hashing	10	B01, B03, R1, V1
47-51	Priority Queue- Implementation of Priority Queue using Heap, Application of Priority Queue		
52-60	Basic concepts of OOPS: Classes, objects and Constructors, Abstraction and Encapsulation, Inheritance	10	B02, V3, V4, V5
61-65	Polymorphism, Message Passing		
66-70	Design the solution of Problem using OOPs paradigm		
71-75	Advance Array Algorithms: Frequency, Prefix Arrays and two pointers, Kadane's Algorithm,	10	B03, B05, R2
76-79	Prime Number, Number Theory: Sieve of Eratosthenes	20	B04, R3, R4
80-83	Divide and Conquer approach, Algorithms and Application		B02, V2
84-86	Greedy Approach with algorithms		B02, V1
87-91	Advanced Dynamic Programming Problems		B02, V3
90-94	Graphs: Introduction, Cycle Detection, Topological Sorting	20	B04, V4
95-100	Graphs: Minimum Spanning Trees, Minimum Spanning Trees Kruskal algorithm, Minimum Spanning Trees Prims algorithm		B01, B05, R2, R4
101-105	Graphs: Bellman Fords Algorithm, Dijkstra Algorithm, Floyd-Warshall Algorithm		B01, B05, R1

10. Industry Interventions:

- The students will receive resources on the LMS system, and their progress will be tracked on the platform based on their submissions (number of submissions, errors encountered, and optimized solutions).

11. Action plan for different types of learners

Slow Learners	Average Learners	Advanced Learners
Remedial Classes	Workshops / Practice Assignment	Advanced Problems

12. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment (Offline/ Online)
Internal Component	Sessional Tests (STs)	02*	40%	Online**
External Component	End Term Examination (ETE)	01	60%	Online**
Total		**100%**		

* Out of the two, the best 01 will be considered.
** Proctored examination will be conducted on Testpad platform.

13. Details of Evaluation Components:

Evaluation Component	Description	Syllabus Covered (%)	Timeline of Examination	Weightage (%)
Internal Component	ST 01	Upto 40% (Lectures 1-42)	Week 7	40
	ST 02	41% - 80% (Lectures 43-84)	Week 12	
External Component	End Term Examination*	100%	As Notified by the Exam Cell	60
Total				100

* Minimum 75% attendance is required to become eligible for appearing in the End Term Examination

14. Format of Evaluation Components:

Type of Assessment	Total Marks	1 Mark MCQ	2 Marks MCQ	5 Marks Coding Question	10 Marks Coding Question
Sessional Tests	40	10	5	2	1
End Term Examination	60	10	10	4	1

15. Revision (if any):

Academic Yearof Previous Version	2024-2025	**Percentage of Revision**	13%
Topics:			

Added:

1. Z-Value Algorithm
2. Prime Number, Number Theory: Sieve of Eratosthenes
3. Floyd-Warshall Algorithm
4. Design and Analysis of Recursive and Backtracking Algorithms

Removed:

1. Index Mapping, Open Addressing, Separate Chaining for Collision Handling,
2. Recursion on Matrics

16. This Document is:

Designation	Name	Signature
Prepared by Course Coordinator	Dr. Osho Sharma	
Verified by Assistant Dean	Dr. Ashutosh Kumar Dubey | Dr. Ravita Chahar	
Date	January 10, 2026	